

ContactMe Android App Free / No-Card Implementation Plan

Firebase Spark + Firestore signaling + Pure WebRTC + Free STUN/TURN + Cloudflare Worker based notification plan

Project: ContactMe

Goal: WhatsApp-like Android personal beta without paid Firebase Blaze or credit card dependency

Scope: Step 82-100 corrected roadmap

Theme: #a605e6

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Date: 20 June 2026

1. Executive Decision

Final direction এই plan-এ Firebase Blaze Plan, Firebase Cloud Functions, paid server, paid TURN, এবং ZegoCloud mandatory dependency বাদ দেওয়া হয়েছে। ContactMe app প্রথমে pure WebRTC + Firestore signaling দিয়ে free/no-card ভিত্তিতে তৈরি হবে।

Card ছাড়া feasible core: Login, user status, real-time messaging, Firestore-based call signaling, STUN/TURN-backed WebRTC audio/video foundation, foreground/background listener, and later Cloudflare Worker + FCM notification bridge.

Important limitation: App fully killed অবস্থায় WhatsApp-like reliable incoming call ringing শুধু Android client + Firestore listener দিয়ে 100% dependable হবে না। Killed-app notification-এর জন্য trusted server side FCM sender দরকার। Card ছাড়া free option হিসেবে Cloudflare Worker ব্যবহার করা হবে।

2. Free / No-Card Technology Stack

Layer	Free/No-Card Choice	Use	Notes
Auth	Firebase Authentication	User login/register	Spark plan suitable for basic auth.
Database	Cloud Firestore	Messages, users, call signaling	Daily free quota; billing needed only if quota beyond free tier is required.
Media	Pure WebRTC	Audio/video media	No third-party call SDK required.
STUN	Google STUN	Direct peer-to-peer NAT discovery	Use <code>stun:stun.l.google.com:19302</code> .
TURN	Metered.ca free TURN	Relay when direct connection fails	Free quota limited; use for dev/testing and small personal beta.
Notifications	FCM + Cloudflare Worker Free	Message/call push fanout	Keep FCM service account secret in Worker, never in APK.
Background	Foreground Service + Firestore listener	Incoming call/message detection while app is alive/backgrounded	Killed-app still needs FCM push.

3. What Changed from the Previous Roadmap

Old item	Problem	Corrected free/no-card replacement
Firebase Cloud Functions	Usually needs Blaze/billing setup for deployment	Cloudflare Worker Free for trusted FCM fanout
ZegoCloud mandatory calling	Third-party SDK/credential dependency and not aligned with pure free WebRTC learning path	Pure WebRTC audio/video with Firestore signaling
Client-side FCM HTTP v1 from Android	Unsafe because service credentials cannot be stored in APK	Android calls Cloudflare Worker; Worker sends FCM securely
Paid TURN server	Credit card/payment dependency	Metered.ca free TURN for testing/ personal beta, later own coturn or paid plan if needed

4. Target Architecture

Android App A App B ----- -----	Firestore / Free Services	Android
Login/Register	-----> Firebase Auth	Login/
Register		
User Profile	<-----> Firestore /users/{uid}	User
Profile		
Message Send	-----> Firestore /chats/...	Listener
receives		
Call Offer	-----> Firestore /calls/{callId}	Incoming
call listener		
ICE Candidate	<-----> Firestore candidate subcollections	ICE
Candidate		
Media	<===== WebRTC STUN/TURN connection	Media
Push Request	-----> Cloudflare Worker Free	FCM
notification		

5. Firestore Data Model

5.1 Users

users/{uid}
- displayName: string
- photoUrl: string?
- fcmToken: string? // optional, for later FCM

```
- online: boolean
- lastSeen: timestamp
```

5.2 Calls

```
calls/{callId}
- callerId: string
- receiverId: string
- type: "audio" | "video"
- status: "ringing" | "accepted" | "rejected" | "ended" | "missed" | "cancelled" | "timeout" | "busy"
- offer: object?
- answer: object?
- createdAt: timestamp
- acceptedAt: timestamp?
- endedAt: timestamp?
```

5.3 ICE Candidates

```
calls/{callId}/callerCandidates/{candidateId}
- sdpMid: string?
- sdpMLineIndex: number?
- candidate: string
- createdAt: timestamp

calls/{callId}/receiverCandidates/{candidateId}
- sdpMid: string?
- sdpMLineIndex: number?
- candidate: string
- createdAt: timestamp
```

6. Step 82 - Free WebRTC Foundation

Scope Step 82 full audio/video call screen তৈরি করবে না। এটি শুধু future call feature-এর foundation তৈরি করবে। এতে limit কম খরচ হবে এবং architecture clean থাকবে।

Step 82 Deliverables

- WebRTC dependency add/verify.
- Call type/status model or enum.
- Call session data class.
- ICE candidate data class.
- Firestore call signaling repository.
- ICE server provider with Google STUN and Metered TURN placeholder.
- Android permissions review.
- docs/learning Step 82 explanation.
- Roadmap/checkpoint doc update.
- Commit and push workflow.

Repository Methods

```
createCallOffer(receiverId, type, offer)
listenForIncomingCalls(currentUserId)
listenToCall(callId)
acceptCallWithAnswer(callId, answer)
rejectCall(callId)
endCall(callId)
addCallerIceCandidate(callId, candidate)
addReceiverIceCandidate(callId, candidate)
listenCallerIceCandidates(callId)
listenReceiverIceCandidates(callId)
```

ICE Server Provider

```
STUN:
stun:stun.l.google.com:19302

TURN placeholder:
turn:YOUR_METERED_TURN_HOST:443?transport=tcp
username: from secure local config / BuildConfig
password: from secure local config / BuildConfig
```

Secret rule: TURN username/password, Firebase service account, API keys that grant server privileges - এগুলো Kotlin source, public docs, GitHub repo, APK, screenshot, বা learning file-এ hardcode করা যাবে না।

7. Manual Preparation Before Giving Step 82 Prompt to Codex

1. Metered.ca account খুলে TURN host, username, password, port, transport note করো.
2. Project repo clean করো: .gradle/, build/, app/build/, APK/AAB/log/screenshots/large outputs Codex context থেকে avoid করো.
3. Firestore connected কিনা confirm করো. তুমি বলেছো already connected, তাই Codex শুধু existing pattern follow করবে.
4. Current branch confirm করো: feature/auth-build.
5. Latest checkpoint doc ready রাখো: docs/28-current-state-and-next-roadmap.md.
6. Codex-কে first response-এ only plan দিতে বলো, edit/build/test নয়.

8. Codex Prompt - Step 82

ContactMe project - Step 82 implementation.

Goal:

Implement the free/no-card WebRTC calling foundation using Firebase Auth + Firestore signaling + WebRTC ICE server configuration skeleton.

Architecture:

- Auth: Firebase Auth already connected.
- User status: Firestore /users/{uid}
 - displayName
 - photoUrl
 - fcmToken optional
 - online
 - lastSeen
- Call signaling: Firestore /calls/{callId}
 - callerId
 - receiverId
 - type: audio/video
 - status: ringing/accepted/rejected/ended/missed/cancelled/timeout/busy
 - offer
 - answer
 - createdAt
 - acceptedAt optional
 - endedAt optional
- ICE candidates:
 - /calls/{callId}/callerCandidates/{candidateId}
 - /calls/{callId}/receiverCandidates/{candidateId}
- Media: Pure WebRTC.
- NAT: Google STUN + Metered.ca TURN free placeholder.
- Background: Foreground Service + Firestore listener later.
- Killed app: later Cloudflare Worker + FCM, not in this step.

Step 82 scope:

1. Add/verify WebRTC dependency if missing.
2. Create call signaling data models.
3. Create call status/type constants or enums.
4. Create Firestore call signaling repository methods:
 - createCallOffer
 - listenForIncomingCalls
 - listenToCall
 - acceptCallWithAnswer
 - rejectCall
 - endCall
 - addCallerIceCandidate
 - addReceiverIceCandidate
 - listenCallerIceCandidates
 - listenReceiverIceCandidates
5. Create ICE server configuration provider/skeleton:
 - Google STUN: stun:stun.l.google.com:19302
 - Metered TURN placeholder only
6. Do not implement full audio/video UI yet.
7. Do not implement ZegoCloud.
8. Do not implement Firebase Cloud Functions.
9. Do not send FCM HTTP v1 directly from Android client.
10. Update docs/learning with Step 82 explanation.
11. Update current roadmap/checkpoint doc.
12. Prepare commit and push after confirmation.

Security rules:

- Do not hardcode TURN username/password in Kotlin source.
- Use placeholder config or local/BuildConfig based approach.
- Do not expose secrets in docs.

Low-token rules:

- Do not scan the full repository.
- First read only:
 - docs/28-current-state-and-next-roadmap.md
 - latest relevant learning/checkpoint file
 - app/build.gradle or build.gradle.kts
 - AndroidManifest.xml
 - existing Firebase/Auth/Firestore repository files
 - existing messaging repository structure only if needed for pattern matching
- Do not run Gradle build/test unless I explicitly approve.
- Do not read generated folders: .gradle, build, app/build, .idea, captures, APK/AAB/log files.
- Do not refactor unrelated messaging code.
- Keep chat response short.
- Put detailed learning explanation inside docs/learning, not in chat.

Before editing:

1. Summarize current state in 5 bullet points.
2. List exact files you will read.
3. List exact files you will create/modify.
4. Give minimal implementation plan.
5. Wait for my confirmation.

9. Corrected Roadmap - Free / No-Card Plan

Step	Corrected Scope	Output
82	Pure WebRTC + Firestore call signaling foundation	Models, repository, ICE config provider, learning doc, commit
83	Current messaging compile/build checkpoint and accumulated bug fix	Stable messaging baseline
84	Cloudflare Worker Free + FCM fanout plan and implementation	Trusted serverless notification sender without Firebase Functions
85	Notification tap deep-link to specific chat	Open exact chat from notification
86	Offline/network failure, retry, duplicate-send and delivery-state polish	Better messaging reliability
87	Two-phone verification: text/image/document/reply/edit/delete and notifications	Messaging E2E checkpoint
88	WebRTC call engine setup and secure TURN config integration	PeerConnection factory, audio constraints, ICE config
89	Firestore call session/state model finalization	Stable call lifecycle schema
90	Outgoing one-to-one audio call	Caller creates offer and ringing session
91	Incoming audio-call screen - accept/reject	Receiver answer/reject flow
92	Background/killed-app incoming-call notification via Cloudflare Worker + FCM	Push-based incoming call alert
93	Microphone permission, speaker, mute and foreground service	Practical audio call controls
94	One-to-one video call with WebRTC	Camera/video track support
95	Camera permission, camera switch, video on/off	Video call controls
96	Ringing/accepted/rejected/busy/cancelled/ended/timeout synchronization	Call state consistency
97	Missed-call notification and call history	Call log and notification polish
98	Network loss, process restart and call cleanup handling	Reliable cleanup and recovery

Step	Corrected Scope	Output
99	Two physical Android phones E2E test and bug fixing	Personal beta readiness
100	Release APK, Firebase/Cloudflare/TURN configuration guide, docs/learning and final checkpoint	Installable APK and complete guide

10. Step 100 Acceptance Target

Authentication

দুই ফোনের user login/register করতে পারবে এবং user status update হবে।

Messaging

Real-time text/image/document/reply/edit/delete flow two-phone test pass করবে।

Notification

Background notification থেকে নির্দিষ্ট chat খুলবে। Killed-app notification Cloudflare Worker + FCM দিয়ে হবে।

Calling

Audio/video call send, receive, accept/reject, end, missed/history support থাকবে।

11. Cost Reality

Service	Cost target	Risk	Mitigation
Firebase Auth	Free Spark usage	Product limits vary by method	Use email/password for initial personal beta
Firestore	Daily free quota	Quota exhaustion	Optimize listeners, paginate messages, cleanup call docs
Metered TURN	Free limited quota	TURN media can consume quota fast	Use TURN only as fallback; keep calls short during testing
Cloudflare Worker	Free daily request limit	Limit exceeded	Use only for notification fanout, not chat storage
FCM	Free notification delivery	Requires trusted sender	Never send FCM HTTP v1 directly from Android client

12. Development Rules for Codex Limit Saving

- একবারে এক step implement করবে।
- Full repo scan করবে না।
- Build/test command approval ছাড়া চালাবে না।
- Generated/cache folders পড়বে না।
- Learning details docs/learning-এ লিখবে, chat-এ repeat করবে না।
- প্রতিটি step শেষে changed files, learning doc, checkpoint update, commit message দিবে।

13. References

1. Firebase pricing plans and Spark/Blaze overview - Firebase official documentation.
2. Cloud Firestore free quota - Firebase/Google Cloud official documentation.
3. FCM HTTP v1 send flow - Firebase official documentation; server/trusted environment should send messages.
4. Cloudflare Workers limits - Free plan daily request limit.
5. Metered.ca TURN pricing - free monthly TURN quota and no credit card note.